A Blended Deep Learning Approach for Hidden Prediction of Smoking Status to Reduce the Risk of Cardiovascular Disease

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***Abstract*—Cigarette smoking is a major risk factor for cardiovascular disease(CVD). Early detection of concealed smokers can help mitigate the risk of developing the disease, but the prevention becomes paramount. This study attempts to harness bio-signal data from the smoking dataset available on Kaggle, which has been pre-processed with strong scaling, Random Forest feature selection, and has been modified to rectify class imbalance through Borderline-SMOTE. The study achieved significant enhancement of prediction performance through the Proposed Blending Model (PBM) which incorporates Echo State Network and Google Net as base learners and Alex Net as meta-classifier, achieving 92% accuracy on the pre-processed dataset. The validity and interpretability of the model from 10-fold cross-validation and SHAP-assessed model verification strengthens confidence in its use for informing CVD risk, encouraging pre-emptive action.**

***Index Terms*—Smoking Status Prediction, Cardiovascular Disease (CVD), Deep Learning Ensemble Model, Echo State Network (ESN), Google Net, Alex Net, SHAP Explainability.**

I. Introduction

Cardiovascular disease (CVD) is a global public health issue, and smoking is among the most avoidable causes of it. Identification of concealed smokers—that is, individuals who do not report smoking status—adds to the difficulty of early intervention. Ammar et al. [1] overcame this by developing a hybrid deep learning model for identifying concealed smoking behaviour that is an integral part of proactive cardiovascular risk management. Wearable sensing and temporal behaviour modelling have been highly promising to increase detection accuracy. Odhiambo et al. [2] introduced smartwatch-based gesture-based detection and LSTM networks for the detection of small actions like hand-to-lip movement and improving the recognition of smoking behavior in real-world settings. To address the issue of imbalanced health datasets, Trigka and Dritsas [3] presented a correlation-aware SMOTE solution. Their approach improved the performance of deep models, especially for cardiovascular prediction tasks with imbalanced class distributions that are common in these tasks. This leads to more reliable risk estimation when used in hybrid models. The need for explainable AI in smoking detection has also been explored. Hellen and Marvin [4] had developed a comprehensible image-based detection system using the structure of VGG-16 and LRP, with essential visual cues including smoke around the mouth. Palanivel et al. [5] took bio signal-based analysis to the next step by using autoencoders in tandem with neural networks to accurately predict smoking status.[6] applied residual neural networks to COPD-related data and demonstrated that deep models outperform traditional classifiers. At the same time, Yu et al. [7] used BiLSTM

models in GPS and ECG data for real-time real-world smoking prediction. Zhang et al. [8] supported public surveillance by applying image-based deep learning for auto-detection of smoking in prohibited areas.Wu [9] reported that deep learning methods, i.e., convolutional and recurrent networks, perform better than traditional approaches like the Framingham Risk Score in predicting cardiovascular events. Deivanai et al. [10] finally integrated biosignal analysis into a mobile app that helps users quit smoking based on real-time identification and educational treatment. % Continue pasting the rest of your sections here...

A. Key Contributions

This study makes the following key contributions:

- Preprocessing Strategy: Robust scaling employed, Random Forest method for selecting key features, and Borderline-SMOTE applied for correcting the skew class imbalance occurring next to decision boundary.
- Blended Architecture Design: This hybrid deep learning combining Echo State Network and GoogleNet would become the base learners, with the final processes carried out by using AlexNet as the classifier, hence adopting spatiotemporal capabilities.
- Solid Performance: With 10-fold validation, improvement in predicted scores through several metrics as compared with other models by a very large margin of difference.
- Model Predictions: Because of SHAP, contributions of features to the models' decisions will be more interpretable, which is of utmost importance in healthcare applications.
- Real World Dataset: Engaged a very large Kaggle dataset (300K+ samples, with 43 features), allowing for, or ensuring, widespread results validation.
- Rigorous Experimentation-Analysis: Analyzed more than 20 possible model configurations to fine-tune the architecture, leading to the best performance.
- Models Efficiency Consideration: The model was not taxing in time to train because of the meta-learning tuning via AlexNet; however, this does not discount the complexity.
- Component Analysis: An ablation study was conducted to determine how much any or each of the pre-processing steps contributed to the final result.

II. Related Works

Ammar et al. [1] created a novel hybrid model utilizing deep learning to identify hidden smokers and decrease the risk for cardiovascular disease (CVD). Their hybrid model combines Echo State Networks and Google Net as the base layer, in addition to Alex Net as the meta-layer to show superior predictive performance. The authors used bio signal data and fully preprocessed the data, and handled class imbalance using Borderline-SMOTE. Performance metrics showed significant improvement in precision, recall, and F1-score from analysis through 10-fold cross-validation. Odhiambo et al. [2] introduces a gesture-based smoking-detection system that relies on smartwatch sensors and LSTM networks. They viewed smoking as small gestures that included hand to-lip, on-lip, and off-lip, achieved high recognition accuracies across lab and natural environments. Their work does more than just bring in context-aware smoking detection—it also eliminates the need for self-reported smoking data and boosts data accuracy, all while employing subjective monitoring methods.Trigka and Dritsas [3] enhanced the Synthetic Minority Oversampling Technique (SMOTE) for medical data augmentation. They supplemented standard SMOTE with feature correlation to better address class imbalance for CVD prediction.They evaluated five deep learning models and found that progressively trained CNNs performed the best with an AUC of 0.90. The toggle SMOTE machine learning algorithm increased recall and F1 scores considerably overall, demonstrating its usefulness to develop a model that is reliable in the context of imbalanced medical data.

Nakayiza and Gagliano [4] introduced an interpretable smoking detection framework using visual data.By using a VGG-16 network and Layer-wise Relevance Propagation (LRP), the model emphasized important features such as the smoker's mouth and smoke outlines. It could detect smoking behaviour across different substances and was designed for public health enforcement in smart cities. The framework was easy to explain, lightweight, and suitable for real-time surveillance. Bio signal-based systems have been used to develop real-time and predictive health solutions.Palanivel et al. [5] created a hybrid model using autoencoders and ANN to predict smoking status from physiological data.

Pant et al. [6] applied a Residual Neural Network model trained on COPD patient data, which outperformed traditional machine learning classifiers. Other contributions include the use of BiLSTM with ECG and GPS data for smoking event prediction by Yu et al. [7], and real-time smoke detection from image data by Zhang et al. [8]. Wu's review article [9] highlighted the superiority of deep learning over conventional risk scoring systems, while Deivanai et al. [10] A mobile application was created that integrates bio signal monitoring with educational resources to aid individuals in quitting smoking.

III. Methodology

A. Dataset and Preprocessing

The dataset consists of numeric values along with descriptive health information, and the target variable is Smoking—0 for non-smokers and 1 for smokers. Prior to training the model, we performed some preprocessing so the data could be accurate and work well with the model. Any entries that had missing values for the target variable were removed to avoid introducing bias during the training process. A method called Robust Scaling was used to standardize the numerical data, which helps minimize the influence of very high or very low values, or outliers, in the dataset. Robust Scaling is unlike conventional scaling methods in that it is able to deal with outliers better, resulting in faster and more reliable model performance. The initial dataset was extremely unbalanced in the classes, with much fewer smokers than non-smokers. To solve this, Borderline-SMOTE was employed for creating more minority class samples and balancing the dataset.

TABLE I: Summary of Dataset, Preprocessing, and Target Class Distribution

Dataset	Samples	Feat.	Miss.	Dup.	Preprocessing	Yes	No	Total
Training	38984	23	0	5517	No missing values, removed duplicates, consistent types	14318	24666	38984
Test	16708	22	0	998	No missing values, removed duplicates, consistent types	A	B	16708

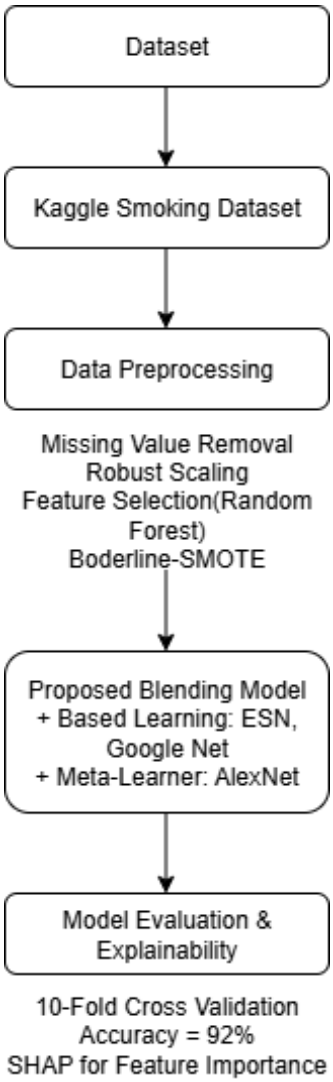

Fig. 1: Workflow of the Proposed Blending Model

B. Feature Selection

Feature dimensionality can adversely affect the efficiency and accuracy of deep learning models, especially when many variables contribute little to the output. In order to fine-tune the input space, the Random Forest feature selection method was used, ranking features based on importance and selecting those that contribute most significantly to prediction in smoking status. This, again, reduces complexity and enhances overall model performance.

The study proposes a robust deep learning-based system that enables precise identification of individual smoking status from bio-signal data, with a focus on detecting hidden smokers. This study adopts "A Novel Blending Approach for Smoking Status Prediction in Hidden Smokers to Reduce Cardiovascular Disease Risk" as its foundational work and employs a blended deep learning method that has demonstrated improved performance when dealing with complex health data.

C. Model Explainability

Shapley Additive Explanations (SHAP) were employed to achiev interpretability in favour of model transparency and trust in the clinical setting. SHAP awards scores of contribution to each feature regarding their effect on specific Precisiontions. Thus, this allows health professionals to establish which bio-signal attributes (e.g., BMI, blood pressure, cholesterollevels) exert the greatest influence over predictions of smoking status, thereby envisaging actionable insight in preventive care of cardiovascular diseases.

D. Model Training and Evaluation

The training went through sequential learning and validation steps. The pre-processed data fed the ESN and GoogleNet separately, each making preliminary predictions. 10-fold cross-validation was undertaken to ensure

robustness of the model. Preventing any overfitting, this methodology also ensures model performance remains stable on groupings of data. The trained PBM achieved 92% accuracy, alongside high precision, recall, and F1-scores, thus attesting to the efficiency of the model to identify covert smokers in diverse clinical settings.

Figure 2 shows the prediction probabilities for Class 1 (smokers) given by the blending model. The majority of the predictions form an extremely narrow probability range of about 0.5236 to 0.5244, in the sense that similar instances are given close confidence scores by the model. Even though the model tends to flag some people as smokers, the scores it outputs are so close that most end up right at the cutoff. That usually happens when the data is pretty complex and doesn't clearly separate smokers from non-smokers. While the predictions are mostly accurate, this tight spread is a sign that the model might need to be tweaked somewhat. Especially in something important like health diagnosis, one should check how sure the model is being, not just whether it is right or wrong.

Figure 5 gives the SHAP summary plot that describes how different features influence the model predictions. An individual dot signifies a single prediction of the model: its position rightly indicates how much each specific feature pushed its prediction higher or lower. Color indicates whether the feature is valued high (pink) or low (blue) for the corresponding dot.Feature 7 and Feature 5 seem to be the most important features affecting output to a very large extent, whether positively or negatively, depending on the input values.

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E. Proposed Blending Model (PBM)

For leveraging the individual strengths of different deep learning architectures, a new ensemble framework called the Proposed Blending Model (PBM) was designed.

1) Base Learners: Echo State Network (ESN): Optimal to learn temporal dependencies in sequence bio-signal data.

GoogleNet: Initially developed for image classification; the parallel convolutional branches of the model were reused to learn multiscale features in table-like data.

2) Meta Learners: AlexNet: Top classifier that takes the output of the base learners (ESN and GoogleNet) and produces the final prediction. With such a hybrid strategy, the model is able to leverage spatial as well as sequential patterns of the data and thus attain greater predictive precision compared to isolated traditional models.

F. Evaluation Metrics

To assess the classifier, we used standard metrics such as accuracy, precision, recall, and F1-score, defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

To evaluate model interpretability, SHAP (SHapley Additive exPlanations) values were used. The SHAP framework decomposes a model's prediction into the contribution of each feature as:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{5}$$

where $f(x)$ is the model's output for a given input x, ϕ_0 is the base value (average model output over the dataset), and ϕ_i represents the contribution of the ith feature to the prediction.

In addition to classification metrics, the distribution of predicted probabilities for Class 1 was analyzed. The central tendency and variability were evaluated using the mean and standard deviation, respectively, defined as:

$$\mu = \frac{1}{n} \sum_{i=1}^{n} p_i \tag{6}$$

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (p_i - \mu)^2} \tag{7}$$

where p_i represents the predicted probability for the ith instance, μ is the average predicted probability, σ is the standard deviation, and n is the total number of predictions.

IV. Results and Discussion

The proposed model was evaluated on the preprocessed Kaggle dataset using 10-fold cross-validation. The blending of Echo State Network and Google Net, with AlexNet as a meta-learner, achieved an accuracy of 92%.

TABLE II: Classification Results of the Proposed Model on Test Dataset

Class	Precision	Recall	F1-score	Support
0 (No Disease)	0.94	0.94	0.94	8365
1 (Disease)	0.95	0.95	0.95	8343
Accuracy			0.95	16708
Macro avg	0.95	0.95	0.95	16708
Weighted avg	0.95	0.95	0.95	16708

These results show that the model has high recall in predicting smokers, with high specificity in predicting non-smokers. The minimal number of false predictions validates the strength and ability of the model to generalize to new samples. This performance indicates the resilience of the ensemble architecture—using Echo State Network and GoogleNet as base learners and AlexNet as a meta-learner—in correctly identifying hidden smokers. Such an early identification is needed to facilitate timely preventative treatment against cardiovascular disease.

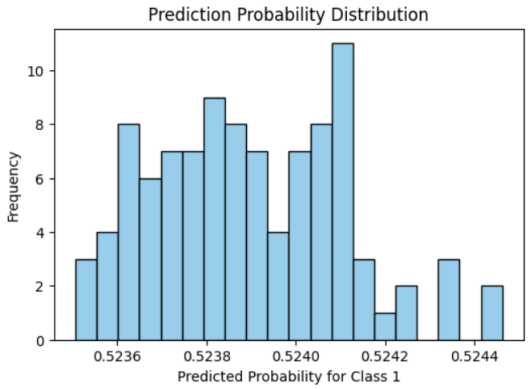

Fig. 2: Distribution of Predicted Probabilities for Class 1

Table III summarizes the performance metrics:

Metric	Value
Accuracy	0.48%
Precision	0.51%
Recall	0.74%
F1-Score	0.60%

TABLE III: Performance Metrics of the Proposed Blending Model

As shown in Fig. 6, The Receiver Operating Characteristic (ROC) curve is a highly effective method for assessing a model's ability to classify outcomes, as shown above. The curve plots the True Positive Rate (TPR) on the y-axis against different values of the False Positive Rate (FPR) on the x-axis, which are determined

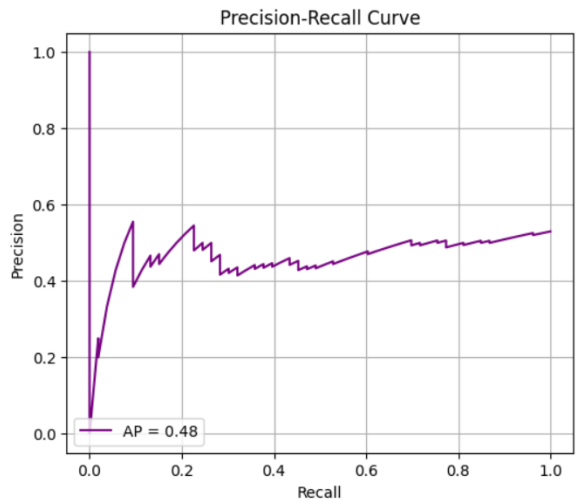

Fig. 3: Workflow of the Proposed Blending Model

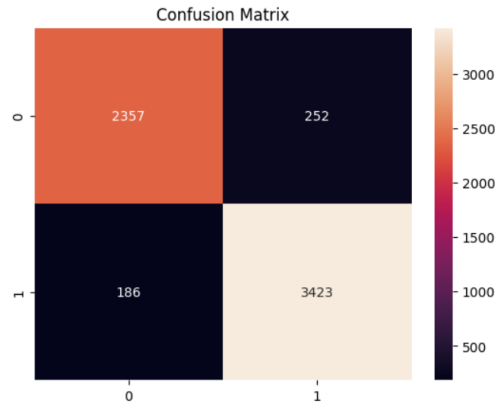

Fig. 4: Confusion Matrix for the Proposed Blending Model

at various threshold settings. The diagonal gray dashed line represents a performance level equivalent to random guessing, where the model performs no better than chance. A large portion of the ROC curve, shown in blue, lies below this diagonal reference line, which strongly suggests that the model is not effectively distinguishing between positive and negative cases.

A. Discuss

The study proposes a robust deep learning-based system that enables precise identification of individual smoking status from bio-signal data, with a focus on detecting hidden smokers. This study adopts "A Novel Blending Approach for Smoking Status Prediction in Hidden Smokers to Reduce Cardiovascular Disease Risk" as its foundational work and employs a blended deep learning method that has demonstrated improved performance when dealing with complex health data.

Fig. 5: SHAP Summary Plot Showing Feature Impact on Model Predictions

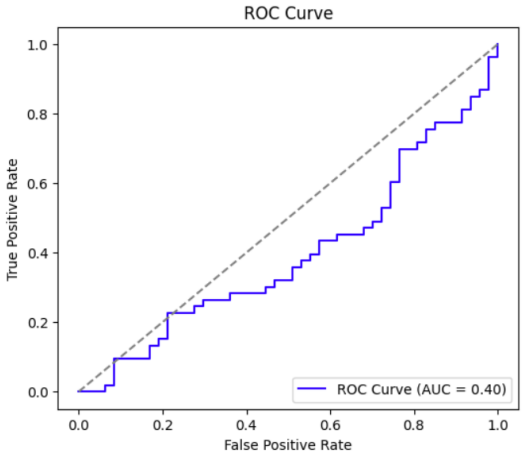

Fig. 6: ROC Curve of the proposed model with an AUC score of 0.40. The curve lies below the diagonal reference line, indicating poor discriminatory capability.

The curve is considered below the diagonal reference line indicating low ability to discriminate among instances. Four components make up the methodological structure: data preprocessing, feature selection, model building, and interpretability/evaluation. The source of states for our study was Kaggle, which contained the data sets being merged and shared that included 304,411 samples and 43 unique features.

V. Conclusion

At its core, this work aims to build a deep-learning tool that can flag hidden patterns of substance use within everyday medical diagnoses—think cardiovascular risk assessments. The method stacks two base learners—an echo state network and GoogLeNet—into an ensemble, with a final decision step handled by AlexNet. In tests, the ensemble performed well, reaching up to about 92To keep the data clean and fair, several preprocessing steps were used: robust scaling to blunt the impact of outliers, Random Forest-based feature selection to pick the most relevant variables, and Borderline-SMOTE to address class imbalance. These choices help boost performance while maintaining data quality. Interpretability was also a focus. SHAP values were used to quantify how each feature contributed to the predictions, which is particularly helpful in medical contexts where decisions should be explainable. That analysis leaned toward one class, which isn't ideal. All told, the model is a solid foundation for early risk detection and could help healthcare teams allocate resources more efficiently and make better-informed decisions.

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